KITABU QUEST

S2

Physics

Mascot: mountain gorilla
Scene: learners exploring physics with simple apparatus and a mountain gorilla mascot
CBC aligned draft learner book

Think • Explore • Achieve

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Title Page

KITABU QUEST S2 Physics

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S2 learners study Physics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use measurements, diagrams, models, and step-by-step reasoning about systems and forces.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

As you finish each topic, return to the success criteria and correct one answer before moving on.

Sources of errors in measurement of physical quantities: Lesson Opener

Learning focus: Sources of errors in measurement of physical quantities

Country link: a safe Rwandan observation, model, data table, or investigation for Sources of errors in measurement of physical quantities.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- identify and explain sources of error in measurements and report

Inquiry questions:
- How can sources of errors in measurement of physical quantities help you solve a real problem in Physics?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Sources of errors in measurement of physical quantities: Learn

Sources of errors in measurement of physical quantities begins with safe behaviour and accurate measurement.

Core idea:
- Physics uses measurements such as length, mass, time, temperature, and volume.
- A measurement is useful only when the number and unit are both written.
- Safety rules protect learners, equipment, and results.
- Read scales at eye level and record values immediately.

Local link: in Rwanda, careful measurement helps in health, building, farming, weather records, transport, and school laboratory work.

Key words:
- Sources of errors in measurement of physical quantities
- Sources
- errors
- measurement
- physical
- quantities
- concept
- observation

Sources of errors in measurement of physical quantities: Worked Example

Investigation model for Sources of errors in measurement of physical quantities: Measure the length of a desk safely.

1. Choose a metre rule or tape measure.
2. Place zero at one end of the desk.
3. Read the scale at eye level.
4. Record the value with the unit, for example 120 cm.
5. Repeat once and compare readings.

Conclusion: a useful measurement includes a number, a unit, and a careful method.

Sources of errors in measurement of physical quantities: Vocabulary And Study Skill

Use these words and study moves before you practise Sources of errors in measurement of physical quantities. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Sources of errors in measurement of physical quantities
- Sources
- errors
- measurement
- physical
- quantities
- concept
- observation

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Sources of errors in measurement of physical quantities without a clear example, method, or evidence.

Sources of errors in measurement of physical quantities: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Sources of errors in measurement of physical quantities.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Sources of errors in measurement of physical quantities: Practice And Correction

Practice for Sources of errors in measurement of physical quantities:

Main outcome: identify and explain sources of error in measurements and report.

1. Name three laboratory safety rules.
2. Measure one classroom object and record the value with a unit.
3. Explain one error that can make a measurement unreliable.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Sources of errors in measurement of physical quantities: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Sources of errors in measurement of physical quantities.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Sources of errors in measurement of physical quantities: Outcome Check

Outcome check for Sources of errors in measurement of physical quantities: Identify and explain sources of error in measurements and report.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Quantitative analysis of linear motion: Lesson Opener

Learning focus: Quantitative analysis of linear motion

Country link: a safe Rwandan observation, model, data table, or investigation for Quantitative analysis of linear motion.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- analyse and solve problems related to linear motion

Inquiry questions:
- How can quantitative analysis of linear motion help you solve a real problem in Physics?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Quantitative analysis of linear motion: Learn

Quantitative analysis of linear motion explains how position changes with time.

Core idea:
- Motion means an object changes position compared with a reference point.
- Distance tells how far an object moves.
- Time tells how long the motion takes.
- Speed compares distance and time.
- A distance-time table or graph can show whether motion is slow, fast, or changing.

Key words:
- Quantitative analysis of linear motion
- Quantitative
- analysis
- linear
- motion
- concept
- observation
- evidence

Quantitative analysis of linear motion: Worked Example

Worked example for Quantitative analysis of linear motion: A cart moves 12 metres in 4 seconds. Find its speed.

1. Speed = distance / time.
2. Distance = 12 m and time = 4 s.
3. 12 / 4 = 3.

Answer: the cart moves at 3 m/s.

Quantitative analysis of linear motion: Vocabulary And Study Skill

Use these words and study moves before you practise Quantitative analysis of linear motion. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Quantitative analysis of linear motion
- Quantitative
- analysis
- linear
- motion
- concept
- observation
- evidence

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Quantitative analysis of linear motion without a clear example, method, or evidence.

Quantitative analysis of linear motion: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Quantitative analysis of linear motion.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Quantitative analysis of linear motion: Practice And Correction

Practice for Quantitative analysis of linear motion:

Main outcome: analyse and solve problems related to linear motion.

1. A bicycle moves 30 m in 6 s. Calculate its speed.
2. Draw a two-row table for distance and time.
3. Explain what happens to speed if the same distance takes more time.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Quantitative analysis of linear motion: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Quantitative analysis of linear motion.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Quantitative analysis of linear motion: Outcome Check

Outcome check for Quantitative analysis of linear motion: Analyse and solve problems related to linear motion.

1. Calculate the speed of an object that moves 20 m in 5 s.
2. State the formula used.
3. Write the unit.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Friction Force: Lesson Opener

Learning focus: Friction Force

Country link: a safe Rwandan observation, model, data table, or investigation for Friction Force.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- explain the effects of a force and its importance in life

Inquiry questions:
- How can friction force help you solve a real problem in Physics?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Friction Force: Learn

Friction Force explains pushes, pulls, and changes in motion.

Core idea:
- A force can start motion, stop motion, change speed, change direction, or change shape.
- Forces are measured in newtons.
- Balanced forces do not change motion.
- Unbalanced forces can make an object accelerate, slow down, or turn.

Key words:
- Friction Force
- Friction
- Force
- concept
- observation
- evidence
- safety
- conclusion

Friction Force: Worked Example

Worked example for Friction Force: A learner pushes a box, and the box starts moving.

Observation: the push is an unbalanced force.

Explanation:
1. The box was at rest.
2. The push acted on the box.
3. The box changed its motion, so the force had an effect.

Conclusion: forces can change the motion of objects.

Friction Force: Vocabulary And Study Skill

Use these words and study moves before you practise Friction Force. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Friction Force
- Friction
- Force
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Friction Force without a clear example, method, or evidence.

Friction Force: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Friction Force.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Friction Force: Practice And Correction

Practice for Friction Force:

Main outcome: explain the effects of a force and its importance in life.

1. Name two pushes and two pulls from daily life.
2. State one effect of force on a moving object.
3. Draw arrows to show forces on a box being pushed.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Friction Force: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Friction Force.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Friction Force: Outcome Check

Outcome check for Friction Force: Explain the effects of a force and its importance in life.

1. Name one push and one pull.
2. State one effect of an unbalanced force.
3. Draw an arrow to show force direction.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Density and Pressure in Solids and Fluid: Lesson Opener

Learning focus: Density and Pressure in Solids and Fluid

Country link: a safe Rwandan observation, model, data table, or investigation for Density and Pressure in Solids and Fluid.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- define pressure and explain factors affecting it

Inquiry questions:
- How can density and pressure in solids and fluid help you solve a real problem in Physics?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Density and Pressure in Solids and Fluid: Learn

Density and Pressure in Solids and Fluid explains particles in solids, liquids, and gases.

Core idea:
- Matter is made of tiny particles.
- In solids, particles are close and vibrate in fixed positions.
- In liquids, particles are close but can slide past each other.
- In gases, particles are far apart and move freely.
- Heating usually makes particles move faster.

Key words:
- Density and Pressure in Solids and Fluid
- Density
- Pressure
- Solids
- Fluid
- concept
- observation
- evidence

Density and Pressure in Solids and Fluid: Worked Example

Model for Density and Pressure in Solids and Fluid: Compare particles in solids, liquids, and gases.

1. Draw solid particles close together in a fixed pattern.
2. Draw liquid particles close but uneven, able to slide.
3. Draw gas particles far apart.
4. Add arrows to show particle movement.

Conclusion: particle arrangement explains shape, flow, and expansion.

Density and Pressure in Solids and Fluid: Vocabulary And Study Skill

Use these words and study moves before you practise Density and Pressure in Solids and Fluid. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Density and Pressure in Solids and Fluid
- Density
- Pressure
- Solids
- Fluid
- concept
- observation
- evidence

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Density and Pressure in Solids and Fluid without a clear example, method, or evidence.

Density and Pressure in Solids and Fluid: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Density and Pressure in Solids and Fluid.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Density and Pressure in Solids and Fluid: Practice And Correction

Practice for Density and Pressure in Solids and Fluid:

Main outcome: define pressure and explain factors affecting it.

1. Draw particles in a solid, liquid, and gas.
2. Explain why gases fill containers.
3. State what heating does to particle movement.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Density and Pressure in Solids and Fluid: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Density and Pressure in Solids and Fluid.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Density and Pressure in Solids and Fluid: Outcome Check

Outcome check for Density and Pressure in Solids and Fluid: Define pressure and explain factors affecting it.

1. Compare particles in a solid and a gas.
2. Draw a simple particle diagram.
3. Explain what heating changes.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Measuring liquid Pressure with Manometer: Lesson Opener

Learning focus: Measuring liquid Pressure with Manometer

Country link: a safe Rwandan observation, model, data table, or investigation for Measuring liquid Pressure with Manometer.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- explain the working principle of manometer use to measure the pressure in fluids

Inquiry questions:
- How can measuring liquid pressure with manometer help you solve a real problem in Physics?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Measuring liquid Pressure with Manometer: Learn

Measuring liquid Pressure with Manometer explains particles in solids, liquids, and gases.

Core idea:
- Matter is made of tiny particles.
- In solids, particles are close and vibrate in fixed positions.
- In liquids, particles are close but can slide past each other.
- In gases, particles are far apart and move freely.
- Heating usually makes particles move faster.

Key words:
- Measuring liquid Pressure with Manometer
- Measuring
- liquid
- Pressure
- with
- Manometer
- concept
- observation

Measuring liquid Pressure with Manometer: Worked Example

Model for Measuring liquid Pressure with Manometer: Compare particles in solids, liquids, and gases.

1. Draw solid particles close together in a fixed pattern.
2. Draw liquid particles close but uneven, able to slide.
3. Draw gas particles far apart.
4. Add arrows to show particle movement.

Conclusion: particle arrangement explains shape, flow, and expansion.

Measuring liquid Pressure with Manometer: Vocabulary And Study Skill

Use these words and study moves before you practise Measuring liquid Pressure with Manometer. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Measuring liquid Pressure with Manometer
- Measuring
- liquid
- Pressure
- with
- Manometer
- concept
- observation

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Measuring liquid Pressure with Manometer without a clear example, method, or evidence.

Measuring liquid Pressure with Manometer: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Measuring liquid Pressure with Manometer.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Measuring liquid Pressure with Manometer: Practice And Correction

Practice for Measuring liquid Pressure with Manometer:

Main outcome: explain the working principle of manometer use to measure the pressure in fluids.

1. Draw particles in a solid, liquid, and gas.
2. Explain why gases fill containers.
3. State what heating does to particle movement.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Measuring liquid Pressure with Manometer: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Measuring liquid Pressure with Manometer.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Measuring liquid Pressure with Manometer: Outcome Check

Outcome check for Measuring liquid Pressure with Manometer: Explain the working principle of manometer use to measure the pressure in fluids.

1. Compare particles in a solid and a gas.
2. Draw a simple particle diagram.
3. Explain what heating changes.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Application of Pascal's principle: Lesson Opener

Learning focus: Application of Pascal's principle

Country link: a safe Rwandan observation, model, data table, or investigation for Application of Pascal's principle.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- explain transmission of pressure in fluids at rest and describe its applications

Inquiry questions:
- How can application of pascal's principle help you solve a real problem in Physics?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Application of Pascal's principle: Learn

Application of Pascal's principle is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:
- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:
- Application of Pascal's principle
- Application
- Pascal's
- principle
- concept
- observation
- evidence
- safety

Application of Pascal's principle: Worked Example

Investigation model for Application of Pascal's principle: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Application of Pascal's principle: Vocabulary And Study Skill

Use these words and study moves before you practise Application of Pascal's principle. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Application of Pascal's principle
- Application
- Pascal's
- principle
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Application of Pascal's principle without a clear example, method, or evidence.

Application of Pascal's principle: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Application of Pascal's principle.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Application of Pascal's principle: Practice And Correction

Practice for Application of Pascal's principle:

Main outcome: explain transmission of pressure in fluids at rest and describe its applications.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Application of Pascal's principle: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Application of Pascal's principle.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Application of Pascal's principle: Outcome Check

Outcome check for Application of Pascal's principle: Explain transmission of pressure in fluids at rest and describe its applications.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Archimedes principle and atmospheric pressure: Lesson Opener

Learning focus: Archimedes principle and atmospheric pressure

Country link: a safe Rwandan observation, model, data table, or investigation for Archimedes principle and atmospheric pressure.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- determine atmospheric pressure using barometer

Inquiry questions:
- How can archimedes principle and atmospheric pressure help you solve a real problem in Physics?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Archimedes principle and atmospheric pressure: Learn

Archimedes principle and atmospheric pressure is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:
- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:
- Archimedes principle and atmospheric pressure
- Archimedes
- principle
- atmospheric
- pressure
- concept
- observation
- evidence

Archimedes principle and atmospheric pressure: Worked Example

Investigation model for Archimedes principle and atmospheric pressure: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Archimedes principle and atmospheric pressure: Vocabulary And Study Skill

Use these words and study moves before you practise Archimedes principle and atmospheric pressure. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Archimedes principle and atmospheric pressure
- Archimedes
- principle
- atmospheric
- pressure
- concept
- observation
- evidence

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Archimedes principle and atmospheric pressure without a clear example, method, or evidence.

Archimedes principle and atmospheric pressure: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Archimedes principle and atmospheric pressure.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Archimedes principle and atmospheric pressure: Practice And Correction

Practice for Archimedes principle and atmospheric pressure:

Main outcome: determine atmospheric pressure using barometer.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Archimedes principle and atmospheric pressure: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Archimedes principle and atmospheric pressure.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Archimedes principle and atmospheric pressure: Outcome Check

Outcome check for Archimedes principle and atmospheric pressure: Determine atmospheric pressure using barometer.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Work, Power and Energy. (II): Lesson Opener

Learning focus: Work, Power and Energy. (II)

Country link: a safe Rwandan observation, model, data table, or investigation for Work, Power and Energy. (II).

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- learner should relate work, power and energy

Inquiry questions:
- How can work, power and energy. (ii) help you solve a real problem in Physics?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Work, Power and Energy. (II): Learn

Work, Power and Energy. (II) connects work, power, and energy changes.

Core idea:
- Work is done when a force moves an object through a distance.
- Energy is the ability to do work.
- Power describes how quickly work is done or energy is transferred.
- Energy can change form, such as chemical to movement, electrical to light, or potential to kinetic.

Key words:
- Work, Power and Energy. (II)
- Work
- Power
- Energy
- concept
- observation
- evidence
- safety

Work, Power and Energy. (II): Worked Example

Worked example for Work, Power and Energy. (II): A learner lifts a book from the floor to a desk.

1. The learner applies an upward force.
2. The book moves upward through a distance.
3. Work is done on the book.
4. The book gains gravitational potential energy.

Conclusion: work can transfer energy to an object.

Work, Power and Energy. (II): Vocabulary And Study Skill

Use these words and study moves before you practise Work, Power and Energy. (II). Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Work, Power and Energy. (II)
- Work
- Power
- Energy
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Work, Power and Energy. (II) without a clear example, method, or evidence.

Work, Power and Energy. (II): Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Work, Power and Energy. (II).
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Work, Power and Energy. (II): Practice And Correction

Practice for Work, Power and Energy. (II):

Main outcome: learner should relate work, power and energy.

1. Give two examples of energy transformation.
2. Explain when work is done on an object.
3. Compare work and power in one sentence.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Work, Power and Energy. (II): Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Work, Power and Energy. (II).
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Work, Power and Energy. (II): Outcome Check

Outcome check for Work, Power and Energy. (II): Learner should relate work, power and energy.

1. Give one example of energy changing form.
2. Explain when work is done.
3. Compare work and power in one sentence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Conservation of mechanical energy in isolated systems: Lesson Opener

Learning focus: Conservation of mechanical energy in isolated systems

Country link: a safe Rwandan observation, model, data table, or investigation for Conservation of mechanical energy in isolated systems.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- apply the principle of conservation of mechanical energy for isolated system

Inquiry questions:
- How can conservation of mechanical energy in isolated systems help you solve a real problem in Physics?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Conservation of mechanical energy in isolated systems: Learn

Conservation of mechanical energy in isolated systems connects work, power, and energy changes.

Core idea:
- Work is done when a force moves an object through a distance.
- Energy is the ability to do work.
- Power describes how quickly work is done or energy is transferred.
- Energy can change form, such as chemical to movement, electrical to light, or potential to kinetic.

Key words:
- Conservation of mechanical energy in isolated systems
- Conservation
- mechanical
- energy
- isolated
- systems
- concept
- observation

Conservation of mechanical energy in isolated systems: Worked Example

Worked example for Conservation of mechanical energy in isolated systems: A learner lifts a book from the floor to a desk.

1. The learner applies an upward force.
2. The book moves upward through a distance.
3. Work is done on the book.
4. The book gains gravitational potential energy.

Conclusion: work can transfer energy to an object.

Conservation of mechanical energy in isolated systems: Vocabulary And Study Skill

Use these words and study moves before you practise Conservation of mechanical energy in isolated systems. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Conservation of mechanical energy in isolated systems
- Conservation
- mechanical
- energy
- isolated
- systems
- concept
- observation

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Conservation of mechanical energy in isolated systems without a clear example, method, or evidence.

Conservation of mechanical energy in isolated systems: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Conservation of mechanical energy in isolated systems.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Conservation of mechanical energy in isolated systems: Practice And Correction

Practice for Conservation of mechanical energy in isolated systems:

Main outcome: apply the principle of conservation of mechanical energy for isolated system.

1. Give two examples of energy transformation.
2. Explain when work is done on an object.
3. Compare work and power in one sentence.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Conservation of mechanical energy in isolated systems: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Conservation of mechanical energy in isolated systems.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Conservation of mechanical energy in isolated systems: Outcome Check

Outcome check for Conservation of mechanical energy in isolated systems: Apply the principle of conservation of mechanical energy for isolated system.

1. Give one example of energy changing form.
2. Explain when work is done.
3. Compare work and power in one sentence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Gas laws' experiments: Lesson Opener

Learning focus: Gas laws' experiments

Country link: a safe Rwandan observation, model, data table, or investigation for Gas laws' experiments.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- describe and analyse gas laws experiments

Inquiry questions:
- How can gas laws' experiments help you solve a real problem in Physics?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Gas laws' experiments: Learn

Gas laws' experiments connects particle number, amount of substance, mass, volume, pressure, and temperature.

Core idea:
- One mole contains Avogadro's number of particles.
- Molar mass links mass and moles: moles = mass / molar mass.
- Gas laws describe how pressure, volume, and temperature change when other conditions are controlled.
- Always list known values, formula, substitution, unit, and final answer.

Key words:
- Gas laws' experiments
- laws
- experiments
- concept
- observation
- evidence
- safety
- conclusion

Gas laws' experiments: Worked Example

Worked example for Gas laws' experiments: Find moles in 18 g of water. Use molar mass of water = 18 g/mol.

1. Formula: moles = mass / molar mass.
2. Substitute: moles = 18 g / 18 g/mol.
3. Calculate: 1 mol.
4. Unit: mol.

Answer: 18 g of water is 1 mol.

Gas laws' experiments: Vocabulary And Study Skill

Use these words and study moves before you practise Gas laws' experiments. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Gas laws' experiments
- laws
- experiments
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Gas laws' experiments without a clear example, method, or evidence.

Gas laws' experiments: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Gas laws' experiments.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Gas laws' experiments: Practice And Correction

Practice for Gas laws' experiments:

Main outcome: describe and analyse gas laws experiments.

1. Calculate moles from a given mass and molar mass.
2. State one gas-law relationship using pressure, volume, or temperature.
3. Show formula, substitution, answer, unit, and one reasonableness check.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Gas laws' experiments: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Gas laws' experiments.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Gas laws' experiments: Outcome Check

Outcome check for Gas laws' experiments: Describe and analyse gas laws experiments.

1. Calculate moles using mass / molar mass.
2. Write the formula and unit.
3. State one controlled condition in a gas-law example.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Magnetization and Demagnetization: Lesson Opener

Learning focus: Magnetization and Demagnetization

Country link: a safe Rwandan observation, model, data table, or investigation for Magnetization and Demagnetization.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- describe methods of magnetization and demagnetization

Inquiry questions:
- How can magnetization and demagnetization help you solve a real problem in Physics?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Magnetization and Demagnetization: Learn

Magnetization and Demagnetization explains magnetic and non-magnetic materials.

Core idea:
- Magnets attract some materials, especially iron and steel.
- A magnet has north and south poles.
- Like poles repel and unlike poles attract.
- Magnetic force can act without touching the object.

Key words:
- Magnetization and Demagnetization
- Magnetization
- Demagnetization
- concept
- observation
- evidence
- safety
- conclusion

Magnetization and Demagnetization: Worked Example

Investigation model for Magnetization and Demagnetization: Test materials with a magnet.

1. Collect a nail, coin, plastic ruler, paper clip, and wood piece.
2. Bring a magnet near each object.
3. Record attracted or not attracted.
4. Group the materials as magnetic or non-magnetic.

Conclusion: iron and steel objects are usually attracted by magnets.

Magnetization and Demagnetization: Vocabulary And Study Skill

Use these words and study moves before you practise Magnetization and Demagnetization. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Magnetization and Demagnetization
- Magnetization
- Demagnetization
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Magnetization and Demagnetization without a clear example, method, or evidence.

Magnetization and Demagnetization: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Magnetization and Demagnetization.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Magnetization and Demagnetization: Practice And Correction

Practice for Magnetization and Demagnetization:

Main outcome: describe methods of magnetization and demagnetization.

1. Test five safe materials with a magnet.
2. Record attracted/not attracted in a table.
3. State what happens when like poles are brought together.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Magnetization and Demagnetization: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Magnetization and Demagnetization.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Magnetization and Demagnetization: Outcome Check

Outcome check for Magnetization and Demagnetization: Describe methods of magnetization and demagnetization.

1. Name two magnetic materials.
2. State what happens with like poles.
3. Describe a fair magnet test.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Applications of Electrostatic: Lesson Opener

Learning focus: Applications of Electrostatic

Country link: a safe Rwandan observation, model, data table, or investigation for Applications of Electrostatic.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- explain applications static charges

Inquiry questions:
- How can applications of electrostatic help you solve a real problem in Physics?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Applications of Electrostatic: Learn

Applications of Electrostatic explains electric charges at rest.

Core idea:
- Some materials become charged when rubbed.
- Charged objects can attract or repel other charged objects.
- Like charges repel; unlike charges attract.
- Dry materials often show static effects more clearly than wet materials.

Key words:
- Applications of Electrostatic
- Applications
- Electrostatic
- concept
- observation
- evidence
- safety
- conclusion

Applications of Electrostatic: Worked Example

Investigation model for Applications of Electrostatic: Observe static attraction.

1. Rub a plastic ruler with dry cloth.
2. Bring it near small paper pieces.
3. Record what happens without touching the paper first.
4. Repeat after waiting one minute.

Conclusion: rubbing can charge materials and cause attraction.

Applications of Electrostatic: Vocabulary And Study Skill

Use these words and study moves before you practise Applications of Electrostatic. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Applications of Electrostatic
- Applications
- Electrostatic
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Applications of Electrostatic without a clear example, method, or evidence.

Applications of Electrostatic: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Applications of Electrostatic.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Applications of Electrostatic: Practice And Correction

Practice for Applications of Electrostatic:

Main outcome: explain applications static charges.

1. Describe how to charge a plastic ruler safely.
2. State whether like charges attract or repel.
3. Name one situation where static charge is noticed.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Applications of Electrostatic: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Applications of Electrostatic.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Applications of Electrostatic: Outcome Check

Outcome check for Applications of Electrostatic: Explain applications static charges.

1. Describe how rubbing can charge an object.
2. State whether like charges attract or repel.
3. Give one safe classroom observation.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Arrangement of resistors in electric circuit: Lesson Opener

Learning focus: Arrangement of resistors in electric circuit

Country link: a safe Rwandan observation, model, data table, or investigation for Arrangement of resistors in electric circuit.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- describe arrangement of resistors in a simple electric circuit

Inquiry questions:
- How can arrangement of resistors in electric circuit help you solve a real problem in Physics?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Arrangement of resistors in electric circuit: Learn

Arrangement of resistors in electric circuit explains moving electric charge in circuits.

Core idea:
- A complete circuit is needed for current to flow.
- Cells, wires, switches, and bulbs can show current effects safely.
- Electricity can produce light, heat, sound, movement, and magnetism.
- Safety matters: never experiment with mains electricity.

Key words:
- Arrangement of resistors in electric circuit
- Arrangement
- resistors
- electric
- circuit
- concept
- observation
- evidence

Arrangement of resistors in electric circuit: Worked Example

Circuit model for Arrangement of resistors in electric circuit: Light a bulb safely.

1. Use a cell, two wires, a switch, and a small bulb.
2. Connect a complete circuit.
3. Close the switch and observe the bulb.
4. Open the switch and observe again.

Conclusion: current flows only when the circuit is complete.

Arrangement of resistors in electric circuit: Vocabulary And Study Skill

Use these words and study moves before you practise Arrangement of resistors in electric circuit. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Arrangement of resistors in electric circuit
- Arrangement
- resistors
- electric
- circuit
- concept
- observation
- evidence

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Arrangement of resistors in electric circuit without a clear example, method, or evidence.

Arrangement of resistors in electric circuit: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Arrangement of resistors in electric circuit.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Arrangement of resistors in electric circuit: Practice And Correction

Practice for Arrangement of resistors in electric circuit:

Main outcome: describe arrangement of resistors in a simple electric circuit.

1. Draw a complete circuit with a cell, switch, and bulb.
2. Explain why a broken circuit stops current.
3. List two electrical safety rules.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Arrangement of resistors in electric circuit: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Arrangement of resistors in electric circuit.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Arrangement of resistors in electric circuit: Outcome Check

Outcome check for Arrangement of resistors in electric circuit: Describe arrangement of resistors in a simple electric circuit.

1. Draw a complete circuit.
2. Explain why a switch can stop current.
3. Write one electrical safety rule.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Reflection of light in curved mirrors: Lesson Opener

Learning focus: Reflection of light in curved mirrors

Country link: a safe Rwandan observation, model, data table, or investigation for Reflection of light in curved mirrors.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- analyse applications of reflected light

Inquiry questions:
- How can reflection of light in curved mirrors help you solve a real problem in Physics?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Reflection of light in curved mirrors: Learn

Reflection of light in curved mirrors explains how light travels and reflects.

Core idea:
- Light travels in straight lines through a uniform medium.
- Shadows form when an opaque object blocks light.
- Reflection occurs when light bounces from a surface.
- Smooth, shiny surfaces reflect light more regularly than rough surfaces.

Key words:
- Reflection of light in curved mirrors
- Reflection
- light
- curved
- mirrors
- concept
- observation
- evidence

Reflection of light in curved mirrors: Worked Example

Investigation model for Reflection of light in curved mirrors: Show that light travels in straight lines.

1. Make small holes in three cards.
2. Place a torch behind the first card.
3. Align the holes so light reaches a screen.
4. Move one card sideways and observe the screen.

Conclusion: the light path is blocked when the holes are not in a straight line.

Reflection of light in curved mirrors: Vocabulary And Study Skill

Use these words and study moves before you practise Reflection of light in curved mirrors. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Reflection of light in curved mirrors
- Reflection
- light
- curved
- mirrors
- concept
- observation
- evidence

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Reflection of light in curved mirrors without a clear example, method, or evidence.

Reflection of light in curved mirrors: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Reflection of light in curved mirrors.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Reflection of light in curved mirrors: Practice And Correction

Practice for Reflection of light in curved mirrors:

Main outcome: analyse applications of reflected light.

1. Draw a straight ray of light from a torch.
2. Explain how a shadow forms.
3. Name one smooth reflector and one rough surface.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Reflection of light in curved mirrors: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Reflection of light in curved mirrors.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Reflection of light in curved mirrors: Outcome Check

Outcome check for Reflection of light in curved mirrors: Analyse applications of reflected light.

1. Draw a straight light ray.
2. Explain how a shadow forms.
3. Describe reflection from a smooth surface.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Basic electronic components: Lesson Opener

Learning focus: Basic electronic components

Country link: a safe Rwandan observation, model, data table, or investigation for Basic electronic components.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- explain the working principle of basic electronic devices

Inquiry questions:
- How can basic electronic components help you solve a real problem in Physics?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Basic electronic components: Learn

Basic electronic components is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:
- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:
- Basic electronic components
- Basic
- electronic
- components
- concept
- observation
- evidence
- safety

Basic electronic components: Worked Example

Investigation model for Basic electronic components: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Basic electronic components: Vocabulary And Study Skill

Use these words and study moves before you practise Basic electronic components. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Basic electronic components
- Basic
- electronic
- components
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Basic electronic components without a clear example, method, or evidence.

Basic electronic components: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Basic electronic components.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Basic electronic components: Practice And Correction

Practice for Basic electronic components:

Main outcome: explain the working principle of basic electronic devices.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Basic electronic components: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Basic electronic components.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Basic electronic components: Outcome Check

Outcome check for Basic electronic components: Explain the working principle of basic electronic devices.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Sources of errors in measurement of physical quantities: Review Clinic 1

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Sources of errors in measurement of physical quantities that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Quantitative analysis of linear motion: Review Clinic 2

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Quantitative analysis of linear motion that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Friction Force: Review Clinic 3

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Friction Force that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Density and Pressure in Solids and Fluid: Review Clinic 4

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Density and Pressure in Solids and Fluid that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Physics.

- Sources of errors in measurement of physical quantities: Explain this term using an example from the book, your school, or your community.
- Quantitative analysis of linear motion: Explain this term using an example from the book, your school, or your community.
- Friction Force: Explain this term using an example from the book, your school, or your community.
- Density and Pressure in Solids and Fluid: Explain this term using an example from the book, your school, or your community.
- Measuring liquid Pressure with Manometer: Explain this term using an example from the book, your school, or your community.
- Application of Pascal's principle: Explain this term using an example from the book, your school, or your community.
- Archimedes principle and atmospheric pressure: Explain this term using an example from the book, your school, or your community.
- Work, Power and Energy. (II): Explain this term using an example from the book, your school, or your community.
- Conservation of mechanical energy in isolated systems: Explain this term using an example from the book, your school, or your community.
- Gas laws' experiments: Explain this term using an example from the book, your school, or your community.
- Magnetization and Demagnetization: Explain this term using an example from the book, your school, or your community.
- Applications of Electrostatic: Explain this term using an example from the book, your school, or your community.
- Arrangement of resistors in electric circuit: Explain this term using an example from the book, your school, or your community.
- Reflection of light in curved mirrors: Explain this term using an example from the book, your school, or your community.
- Basic electronic components: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Physics answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Physics that shows what you can now do independently.

Your project must include:
1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.